Experiment 10

Aim : Study and Implement containerization using Docker

Theory :

1. What is containerisation

   Containerization is a lightweight virtualization technique where applications and their dependencies are packaged into isolated units called containers.

   Key Points:

   1) A container includes:
      1. Application code
      2. Libraries
      3. Dependencies
   2) Runs on a shared host operating system
   3) Uses tools like Docker

   Advantages:

   1) Fast startup time
   2) Portable (works on any system)
   3) Efficient resource usage
   4) Easy deployment and scaling

2. Difference between containers and VM

Feature	Containers	Virtual Machines
Architecture	Share host OS	Each VM has its own OS
Size	Lightweight (MBs)	Heavy (GBs)
Startup Time	Seconds	Minutes
Performance	Near-native	Slightly slower
Isolation	Process-level	Full OS-level
Resource Usage	Low	High
Tools	Docker, Kubernetes	VMware, VirtualBox

3. Security in containers

   Container security ensures that applications running inside containers are protected from threats.

   Security Features:

   1. Namespaces
      - Isolate processes, network, and filesystem
   2. Control Groups (cgroups)
      - Limit CPU, memory, and resources
   3. Immutable Images
      - Containers use read-only images
   4. Least Privilege
      - Avoid running containers as root
   5. Image Scanning
      - Detect vulnerabilities in images

   Common Security Risks:

   - Vulnerable base images
   - Misconfigured containers
   - Running as root user
   - Exposed ports

Activity :

1. Demonstration of creating, finding, building, installing, and running Linux/Windows application containers inside local machine or cloud platform.

   Steps :

   1. Launch EC2 Instance (Linux)

   Login to AWS Academy Lab

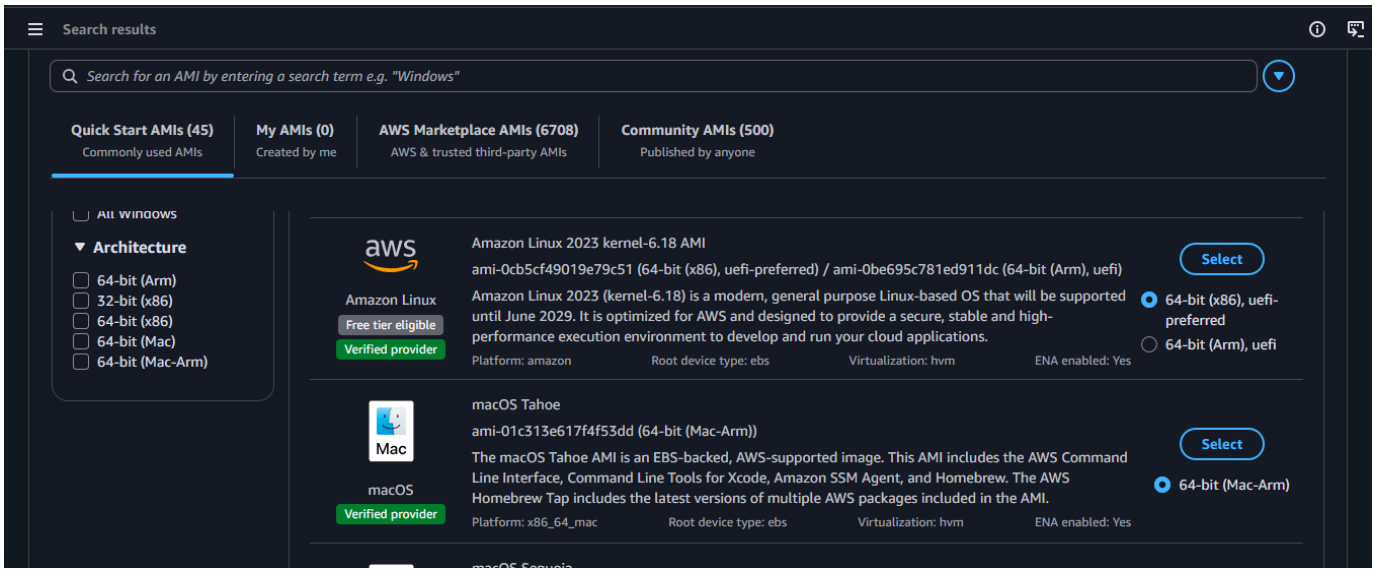
   Go to EC2 → Launch Instance

   Select:

- AMI: Ubuntu (Linux)
- Instance type: t3.micro

  Allow:

- SSH (22)
- HTTP (80)
- Launch and connect via SSH



2. Connect to EC2
3. Install Docker

```
sudo yum update -y
sudo yum install docker -y
sudo systemctl start docker
sudo systemctl enable docker
```

4. Verify Docker

```
docker --version
```

```
  Running scriptlet: docker-25.0.14-1.amzn2023.0.2.x86_64                                                    11/11
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /usr/lib/systemd/system/docker.socket.

  Running scriptlet: container-selinux-4:2.245.0-1.amzn2023.noarch                                           11/11


  Running scriptlet: docker-25.0.14-1.amzn2023.0.2.x86_64                                                    11/11
  Verifying        : container-selinux-4:2.245.0-1.amzn2023.noarch                                            1/11
  Verifying        : containerd-2.2.1-1.amzn2023.0.1.x86_64                                                   2/11
  Verifying        : docker-25.0.14-1.amzn2023.0.2.x86_64                                                     3/11
  Verifying        : iptables-libs-1.8.8-3.amzn2023.0.2.x86_64                                                4/11
  Verifying        : iptables-nft-1.8.8-3.amzn2023.0.2.x86_64                                                 5/11
  Verifying        : libcgroup-3.0-1.amzn2023.0.1.x86_64                                                      6/11
  Verifying        : libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64                                       7/11
  Verifying        : libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64                                                8/11
  Verifying        : libnftnl-1.2.2-2.amzn2023.0.2.x86_64                                                     9/11
  Verifying        : pigz-2.5-1.amzn2023.0.3.x86_64                                                          10/11
  Verifying        : runc-1.3.4-1.amzn2023.0.2.x86_64                                                        11/11

Installed:
  container-selinux-4:2.245.0-1.amzn2023.noarch          containerd-2.2.1-1.amzn2023.0.1.x86_64         docker-25.0.14-1.amzn2023.0.2.x86_64
  iptables-libs-1.8.8-3.amzn2023.0.2.x86_64              iptables-nft-1.8.8-3.amzn2023.0.2.x86_64       libcgroup-3.0-1.amzn2023.0.1.x86_64
  libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64     libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64      libnftnl-1.2.2-2.amzn2023.0.2.x86_64
  pigz-2.5-1.amzn2023.0.3.x86_64                         runc-1.3.4-1.amzn2023.0.2.x86_64

Complete!
[ec2-user@ip-172-31-14-218 ~]$ sudo systemctl start docker
[ec2-user@ip-172-31-14-218 ~]$ sudo systemctl enable docker
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
[ec2-user@ip-172-31-14-218 ~]$ docker --version
Docker version 25.0.14, build 0bab007
[ec2-user@ip-172-31-14-218 ~]$
```

5. Pull Image

```
sudo docker pull ubuntu
```

sudo docker run -it ubuntu

6. Install Software inside Container

apt update

apt-get install python3 -y

7. Create Application

apt-get install nano -y

nano app.py

print("Hello from Container")

python3 app.py

```
print("Hello From Docker")
```

```
aws  Search [Alt+S] Ask Amazon Q  United States (N. Virginia)  Account ID: 6927-3142-6168
voclabs/user4779140=imtiyaz.khan@slrtce.in
done.
root85d1244f8e675:/# nano app.py
bash: nano: command not found
root85d1244f8e675:/# apt-get install nano -y
Reading package lists... Done
Building dependency tree... Done
Buiding state information... Done
Suggested packages:
  hunspell
The following NEW packages will be installed:
  nano
0 upgraded, 1 newly installed, 0 to remove and 10 not upgraded.
Need to get 282 kB of archives.
After this operation, 056 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 nano amd64 7.2-2ubuntu0.1 [282 KB]
Fetched 202 kB in 1s (455 KB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselecked package nano.
(Reading database ... 6208 files and directories currently installed.)
Preparing to unpack .../nano_7.2-2ubuntu0.1_amd64.deb ...
Unpacking nano (7.2-2ubuntu0.1) ...
Setting up nano (7.2-2ubuntu0.1) ...
update-alternatives: using /hin/nano to provide /usr/bin/editor (editor) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/manl/editor.1.gz because associated file /usr/share/man/manl/nano.1.gz (of link group
editor) doesn't exist
update-alternatives: using /hin/nano to provide /usr/bin/pico (pico) in auto mode
update-alternatives: warning: skip creation of /uar/share/man/manl/pico.1.gz because associated file /uar/share/man/manl/nano.1.gz (of link group p
co) doesn't exist
```

Conclusion :

Advantages of containers :

1. Lightweight
   - Containers share the host OS → no need for separate OS
   - Uses less memory and storage
2. Fast Startup
   - Starts in seconds (unlike VMs which take minutes)
3. Portability
   - Runs the same across environments (dev, test, production)
   - Example: Apps in Docker run anywhere

4. Efficient Resource Utilization
   - Multiple containers can run on a single system
   - Better CPU and memory usage
5. Scalability
   - Easy to scale up/down (used in microservices architecture)

Disadvantages of containers :

1. Less Secure than VMs
   - Share host OS → weaker isolation compared to VMs
2. OS Dependency
   - Containers must use the same OS kernel as host
   - Cannot run Windows container on Linux host (directly)
3. Complex Networking
   - Managing communication between containers can be difficult
4. Storage Issues
   - Data persistence is tricky (containers are ephemeral)
5. Management Complexity
   - Large-scale systems need orchestration tools like
     - Kubernetes